CAP 3330

Module 3

Analysis **o**f **Va**riances (ANOVA)

Topics

- ANOVA **VS** t test for two samples.
- ANOVA test for **one factor** (**one-way** ANOVA).

ANOVA
VS
t test to compare two means

- They are both tests to compare the means of two or + populations.
- ANOVA is ***recommended*** to compare three means or more, whereas t test is ***recommended*** only to compare two means.
- ANOVA stands for Analysis of Variances. In contrast to the t test, ANOVA does NOT focus on the difference between the sample means. It focuses on evaluating and comparing multiple types of variances.

Why is ANOVA ***recommended*** to compare **three means or +**, whereas t test is ***recommended*** only to compare **two means**?

How many t tests are needed if you have three groups to compare?

Answer: Three t tests because three groups result in three pairs of means.

How many t tests are needed if you have four groups to compare?

Answer: Six t tests because four groups result in six pairs of means.

As the number of groups to compare increases, the possible number of pairs increases too. *This could have a negative implication!*

In Statistics, every single test that you run has a probability of leading you to the wrong conclusion. The more tests you need to run, the more chances there are of making a wrong conclusion at some point.

Example: t test to compare four means (read independently)

When three or more means are being compared, can we still use the t test to compare them? YES, but… there are risks in doing so. See next:

Example: $\mu_1=\mu_2=\mu_3=\mu_4$?

Comparison	Probability of making type 1 error
$\mu_1 = \mu_2$	α=0.05
$\mu_1 = \mu_3$	α=0.05
$\mu_1 = \mu_4$	α=0.05
$\mu_2 = \mu_3$	α=0.05
$\mu_2 = \mu_4$	α=0.05
$\mu_3 = \mu_4$	α=0.05

Instead of focusing on the probability of making a type 1 error on each individual comparison, a more useful question is: What's the probability of making a type 1 error **in at least one of these six comparisons**?

Easier to compute this probability using the complement:

Probability of not committing type 1 error for one comparison= 1-0.05=0.95

Probability of not committing type 1 error in any of the six comparisons= 0.95^6= 0.7351

Probability of committing type 1 error in at least one comparison= 1-0.7351= **0.2649**

ANOVA terminology

Dependent/ Outcome variable: The quantitative variable under investigation.

Example: Crop yield

Factor: The qualitative variable whose effect on the dependent variable is under investigation.

Example: The type of fertilizer

Factor levels: The labels/categories of the factor.

Example: Domestic fertilizer, foreign fertilizer, and water.

If only one factor is being studied, **the factor levels** are also called **treatments**.

- The t test applies to one factor with two levels.
- One-way ANOVA applies to one factor with 3 or + levels (and later, …., we will study ANOVA with more than one factor).

Example of ANOVA terminology

Examples: The US Golf Association whishes to compare the mean distance traveled by four different brands of golf balls. The following experiment is conducted: 10 balls of each brand are randomly selected. Each ball is struck by a robot, and the distance traveled is recorded.

DV, factor, and levels?

When is ANOVA used?

1. ANOVA applies to both randomized controlled trials and observational studies.

2. In both cases, the groups being compared are assumed to differ only with respect to the factor levels (treatments).

3. Ensuring group comparability is the subject of an extensive branch of Statistics (beyond the scope of this course). In this course, we assume the groups are comparable (comparable in a statistical sense).

Steps to conduct one way ANOVA (1 factor ANOVA)

1- Set up hypotheses

Ho: $\mu_1 = \mu_2 = \ldots = \mu_k$
Ha: At least two of the k means are different

2- Test statistic

F = MST / MSE

MST (Mean Square of Treatment) measures variability between groups.

MSE (Mean Square of Error) measures the variability within groups.

3- Decision rule

P value= Area to the right of the F value obtained in (2)

4- Make decision and conclude

Practice in R Studio!

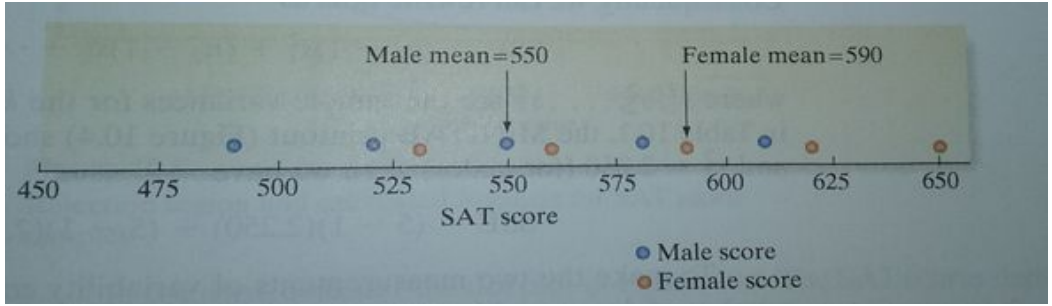
Method of solution of one-way ANOVA

Goal of ANOVA: Compare the means of multiple (>=3) populations or groups.

- ANOVA divides the variability (variation) in the data into two components: **variability within** groups and **variability between** groups.

- A comparison of these two components (the variability within and the variability between) is the basis in ANOVA to conclude whether the means are equal.

For illustration, let's explore an example where only two groups are being compared: the mean SAT score between males and females.



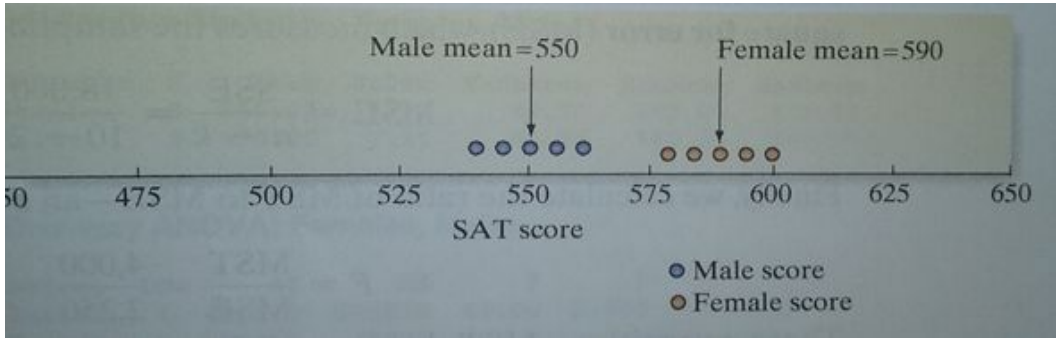
Variability within groups: High

Variability between groups: High

ANOVA conclusion: The means of these two groups are NOT statistically different.

Method of solution of one-way ANOVA

Another scenario with the same previous example: the mean SAT score between males and females.



Variability within groups: Low

Variability between groups: High

ANOVA conclusion: The means of these two groups are statistically different.

Technical details of the solution of one-way ANOVA

SKIP (Review independently if interested)

For k groups (k means under comparison) and nj element within each group:

- Measure of the **variability within groups**: Sum of squares for error (**SSE**).

$$\text{SSE} = \sum_{j=1}^{n_1}(x_{1j} - \bar{x}_1)^2 + \sum_{j=1}^{n_2}(x_{2j} - \bar{x}_2)^2 + \ldots \sum_{j=1}^{n_k}(x_{kj} - \bar{x}_k)^2$$

- Measure of the **variability between groups**: Sum of squares for treatments (**SST**).

$$\text{SST} = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2$$

Technical details of the solution of one-way ANOVA

SKIP (Review independently if interested)

- **SSE** and **SST** are not comparable (explanation beyond scope of course).

- When each sum of square is divided by its corresponding degrees of freedom, they both become a measure of the variability in the DV.

Degrees of freedom for **SSE**= n-k

$$MSE = \frac{SSE}{n - k}$$

MSE: Mean square of the error

Degrees of freedom for **SST**= k-1

$$MST = \frac{SST}{k - 1}$$

MST: Mean square of treatments

Technical details of the solution of one-way ANOVA

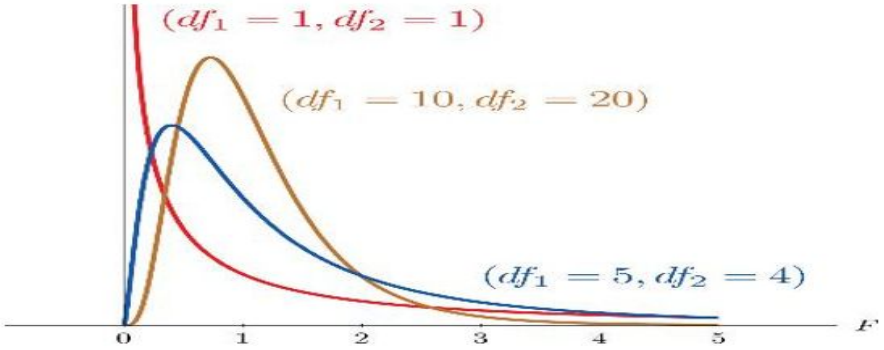
SKIP (Review independently if interested)

- When **MST** is divided by **MSE**, we obtain a statistic with an F distribution

$$\frac{\text{MST}}{\text{MSE}}$$

This ratio follows an F-distribution with $\boldsymbol{\nu_1}$ (numerator DF) = k-1, and $\boldsymbol{\nu_2}$ (denominator DF) = n-k

The F distribution



Practice in R Studio! (cont.)

Practical significance

- We could use Cohen's effect size index for one-way ANOVA

7. One-way analysis of variance

$$f = \frac{\sigma_m}{\sigma}$$

Where, **sigma m** is the standard dev of the sample means and **sigma** is the standard deviation of all observations together.

- A better way is to do a post-hoc test to assess the difference between pairs of means and find out: How many pairs of means show a significant difference? What pair has the largest difference?

Post-hoc tests (pairwise comparisons) in one-way ANOVA

Multiple comparisons of means

- The ANOVA F test provides an overall comparison among the means of all the groups. A rejection of Ho indicates at least two means are different, but we usually want to know more.

- **Post hoc tests** perform all pairwise comparisons **using a modified version of the t test for two samples**.
  - We saw that doing pairwise t tests to compare pairs of means inflates the probability of making type I error.
  - We need to make some ***adjustments*** when making pairwise comparisons so that the probability of making type I error does not increase so much.

Multiple comparisons of means

From a previous slide: Risk of multiple pairwise comparisons using t test. Example with 4 groups (4 means)

Comparison	Probability of making type 1 error	Probability of making type 1 error in at least one of the 6 comparisons
$\mu_1 = \mu_2$	α=0.05	Easier to compute this probability using the complement:
$\mu_1 = \mu_3$	α=0.05	
$\mu_1 = \mu_4$	α=0.05	Probability of not committing type 1 error for one comparison=1-0.05=0.95
$\mu_2 = \mu_3$	α=0.05	
$\mu_2 = \mu_4$	α=0.05	
$\mu_3 = \mu_4$	α=0.05	Probability of not committing type 1 error in any of the six comparisons= 0.95^6= 0.7351 Probability of committing type 1 error in at least one comparison= 1-0.7351= **0.2649**

The *probability of making a type I error in at least one of the pairwise comparisons* (e.g., 0.2649 in this example) is called the family-wise error rate or experiment-wise error rate.

- We denote the experiment-wise error rate by α_w

- α is still the error rate for each pairwise comparison.

- Need to adjust α (or the P value) for each pairwise comparison in order to get a low value of α_w (i.e., smaller α so α_w is reasonably small too).

Post hoc tests that adjust for experiment wise error rate

- Bonferroni
- Tukey
- Holm
- Duncan
- Dunnet

- No best procedure (depends on how many means are being compared, how large we expect the differences between the means to be (= expected effect size))

Bonferroni's method (read independently if interested)

(1) P (at least one Type I error in all comparisons) $\leq \alpha_w$

(2) $\alpha_w = m*\alpha$

m: total number of comparisons

α : significance level for each individual pairwise comparison

(3) From (2), $\alpha = \alpha_w /m$. In short, this method computes the α to use in each individual comparison based on the desired experiment-wise error (α_w).

Equivalently, instead of lowering the value of α as in (3), we keep α with its original value (i.e., say, 0.05) and we multiply the P value of each individual comparison by m. This is what is done in R.

Bonferroni's method (read independently if interested)

Bonferroni's is easy to apply but has very **low statistical power** when **m is large**.

In R: pairwise.t.test (vector with DV, vector with values of factor ,p.adjust.method = "bonferroni")

Tukey's method (read independently if interested)

When sample sizes for all treatments are equal (also applicable when all sample sizes are not the same):

$$(\overline{X}_A - \overline{X}_B) \pm q \sqrt{\frac{MSE}{n_t}}$$

$\overline{X}_A$, $\overline{X}_B$: Sample means for treatments being compared.

q: Value of the studentized range distribution (another probability distribution). q depends on α_w , m and DF of MSE.

nt: Common sample size for all samples.

- More statistical power than Bonferroni's when m is large.
- **In R**: TukeyHSD (object resulting from aov())

t test **VS** ANOVA **VS** Post hoc test (adjusted t test)

Pairwise comparisons using standard (unadjusted) t tets result in a high family-wise error rate.

Pairwise comparisons using adjusted t tets (Bonferroni, Tuket,...) keep family-wise error rate at acceptable levels.

ANOVA is very good at detecting that at least one pair of means are different.

If adjusted pairwise comparisons keep family wise error low, why do we need ANOVA?

In some situations, ANOVA is more powerful (higher chances to detect a true difference) than some pairwise comparisons.

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Analysis of Variances (ANOVA)